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2026 tssm units 3 4 - latest

Chemistry (Xavier College)



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2026 Trial Examination

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Chemistry

Units 3 and 4 – Question and Answer Book

VCE Examination

Reading time: **15 minutes**
Writing time: **2 hours and 30 minutes**

Materials supplied

- Question and Answer Book of 43 pages
- VCE Chemistry Data Book
- Multiple-Choice Answer Sheet

Instructions

- Write your responses in English.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents pages

Section A (30 questions, 30 marks) _____ 2- 16
Section B (7 question, 90 marks) _____ 17 - 42

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SECTION A – Multiple-choice questions

Instructions

- Answer **all** questions in pencil on the Multiple-Choice Answer Sheet.
- Choose the response that is **correct** or that **best answers** the question.
- A correct answer scores 1; and incorrect answer scores 0.
- Marks will **not** be deducted for incorrect answers.
- No mark will be given if more than one answer is chosen.
- Unless otherwise indicated, diagrams in this book are not drawn to scale.

Question 1

Which of the following best describes a method to produce biomethane?

A. The fermentation of sugars obtained from plant mass.

B. The anaerobic digestion of waste by bacteria.

C. The fractional distillation of natural gas.

D. The transesterification of plant oils.

Question 2

A nutrition label on a packet of bread is shown below.

Nutrient	Average quantity (g/100 g)
Carbohydrates	22.0
Fats	1.0
Protein	4.0

The energy content of a 50 g serving of bread is closest to

A. $2.3 \times 10^2\ kJ$

B. $4.6 \times 10^2\ kJ$

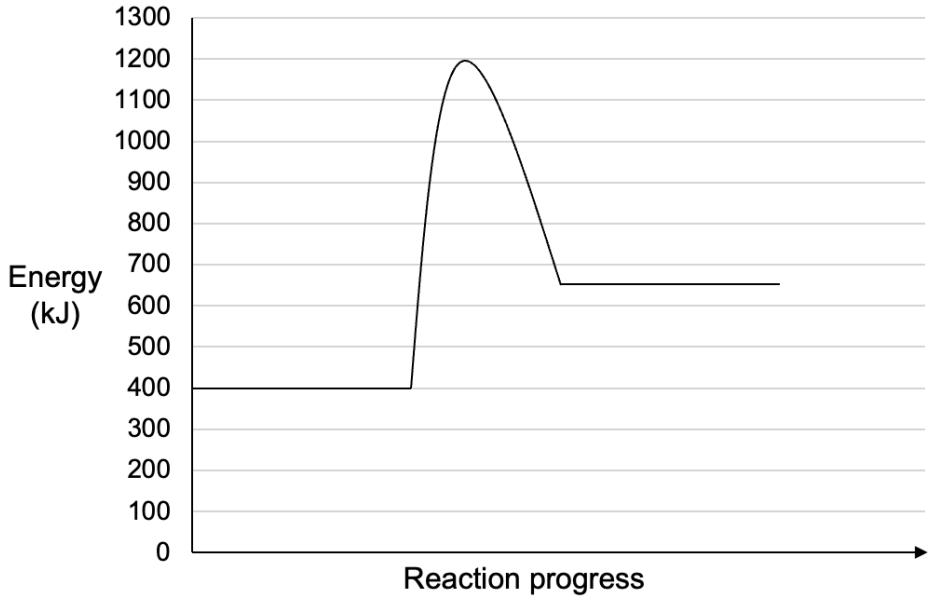
C. $7.0 \times 10^2\ kJ$

D. $9.1 \times 10^2\ kJ$

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Question 3

The energy profile diagram for a particular chemical reaction is shown below.



Consider the reverse reaction.

Which of the following represents the activation energy (E_A), and the change in enthalpy (ΔH), of the reverse reaction?

	E_A (kJ)	ΔH (kJ)
A.	$+800$	$+250$
B.	$+800$	-250
C.	$+550$	$+250$
D.	$+550$	-250

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Question 4

A particular biogas system used in the production of onsite electricity has an efficiency of 40%. A 1.0 kg sample of the biogas produces 8 MJ of useful energy.

Assuming the rest of the of biogas composition is made up of non-combustible gases, the methane content of the gas is closest to

A. 8.6%
B. 22%
C. 36%
D. 86%

Question 5

The expression for the equilibrium constant for a particular reaction is shown below.

$$K = \frac{[NO]^2[O_2]}{[NO_2]^2}$$

The chemical equation that represents the reaction is

A. $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$
B. $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$
C. $4NO(g) + 2O_2(g) \rightleftharpoons 4NO_2(g)$
D. $2NO_2(g) \rightleftharpoons NO(g) + O_2(g)$

Question 6

Which of the following design modifications would not improve the efficiency of a proton-exchange membrane (PEM) ethane-oxygen fuel cell?

A. Increasing the porousness of the cathode material.
B. Increasing the ionic conductivity of the proton-exchange membrane.
C. Increasing the capacity of the ethane fuel tank.
D. Embedding a catalyst in the anode material.

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Question 7

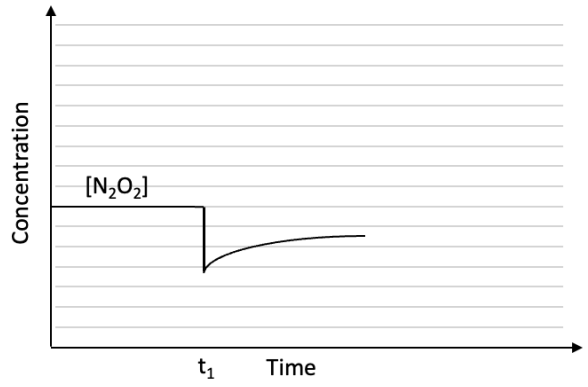
Consider the following equilibrium system.

$$N_2O_2(g) \rightleftharpoons 2NO_2(g) \quad \Delta H > 0$$

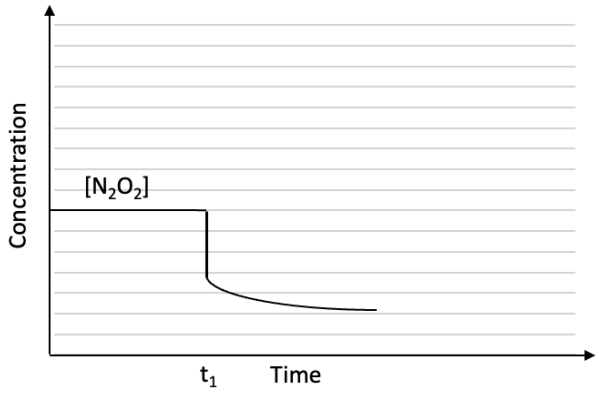
The reaction mixture is at equilibrium, when at time point t_1 the volume of the reaction vessel is doubled.

Which of the following concentration-time graphs best represents the changes in N_2O_2 concentration that occur following the change at t_1?

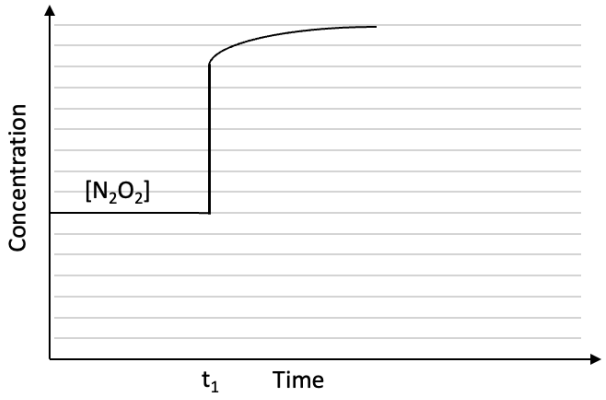
A.



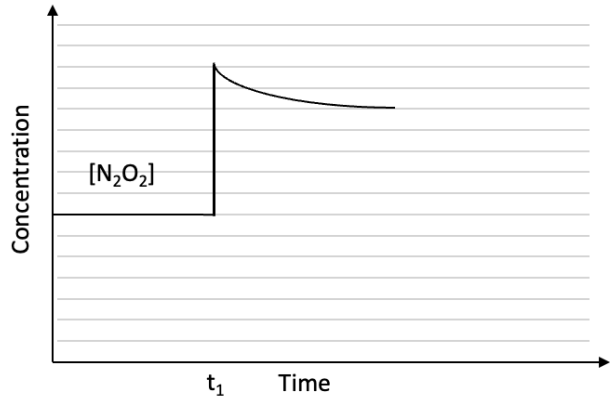
B.



C.



D.



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Use the following information to answer Questions 8 and 9.

A solution calorimeter is calibrated by passing a current of 1.50 A through a heater with a voltage of 8.00 V for 3 minutes. The temperature of the water increased from 23.8°C to 26.4°C.

Question 8

The calibration factor of the calorimeter is closest to

A. 13.8 J °C^{-1}

B. 81.8 J °C^{-1}

C. 277 J °C^{-1}

D. 831 J °C^{-1}

Question 9

Following calibration, a 150.0 g sample of sucrose, $C_{12}H_{22}O_{11}$, is added to the solution calorimeter. The temperature decreases by 1.20°C.

Which of the following represents the enthalpy of dissolution of sucrose determined from this experiment data?

A. $\Delta H = -2.27\ kJ\ mol^{-1}$

B. $\Delta H = +2.27\ kJ\ mol^{-1}$

C. $\Delta H = -2.92\ kJ\ mol^{-1}$

D. $\Delta H = +1.50\ kJ\ mol^{-1}$

Question 10

Which of the following statements is true regarding photosynthesis?

A. In writing a thermochemical equation for photosynthesis, the value of ΔH is $-2840\ kJ$.

B. The amount of energy stored in the bonds within carbon dioxide and water is less than the amount of energy stored in the bonds within glucose and oxygen.

C. The energy required to break the bonds within carbon dioxide and water is less than the energy released when the bonds form within glucose and oxygen.

D. For every 1 mol of carbon dioxide that reacts, 6 mol of glucose is produced.

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Question 11

5.0 g of butane undergoes combustion in a sealed container containing 50 L of air. Assuming the air in the container is made up of 20% oxygen, and reaction occurs at SLC, which of the following statements is correct?

A. Butane is the limiting reactant, and 47 g of unreacted oxygen remains in excess.
B. Butane is the limiting reactant, and 0.60 L of unreacted oxygen remains in excess.
C. Oxygen is the limiting reactant, and 1.4 g of unreacted butane remains in excess.
D. Oxygen is the limiting reactant, and 4.2 g of unreacted butane remains in excess.

Question 12

The overall discharge reaction that occurs in a particular lithium-ion battery is represented by the chemical equation below.

$$4Li(s) + 3CO_2(g) \rightarrow 2Li_2CO_3(s) + C(s)$$

Which of the following represents the half-equation of the reaction that occurs at the anode during discharge?

A. $3CO_2 + 4e^- \rightarrow 2{CO_3}^{2-}(s) + C(s)$
B. $3CO_2 \rightarrow 2{CO_3}^{2-}(s) + C(s) + 4e^-$
C. $Li^+(s) + e^- \rightarrow Li(s)$
D. $Li(s) \rightarrow Li^+(s) + e^-$

Question 13

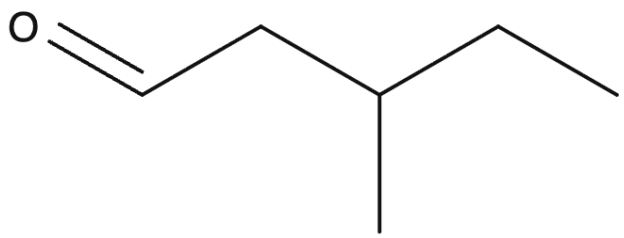
A student sets up an electrolytic cell to plate an inert carbon electrode with nickel. The cell operates for 20.0 minutes, with an average current of 1.80 A.

What is the theoretical mass of nickel that this could produce?

A. 0.02 g
B. 0.66 g
C. 1.30 g
D. 2.61 g

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Question 14



What is the IUPAC name of the molecule shown above?

A. 3-methylpentanal
B. 2-ethylbutanal
C. 3-methylpentan-1-one
D. 3-ethylbutan-1-one

Question 15

A sample containing the amino acids phenylalanine, serine, and valine was analysed using high performance liquid chromatography, with a polar mobile phase and a non-polar stationary phase. Refer to **Item 19 of the Data Book**.

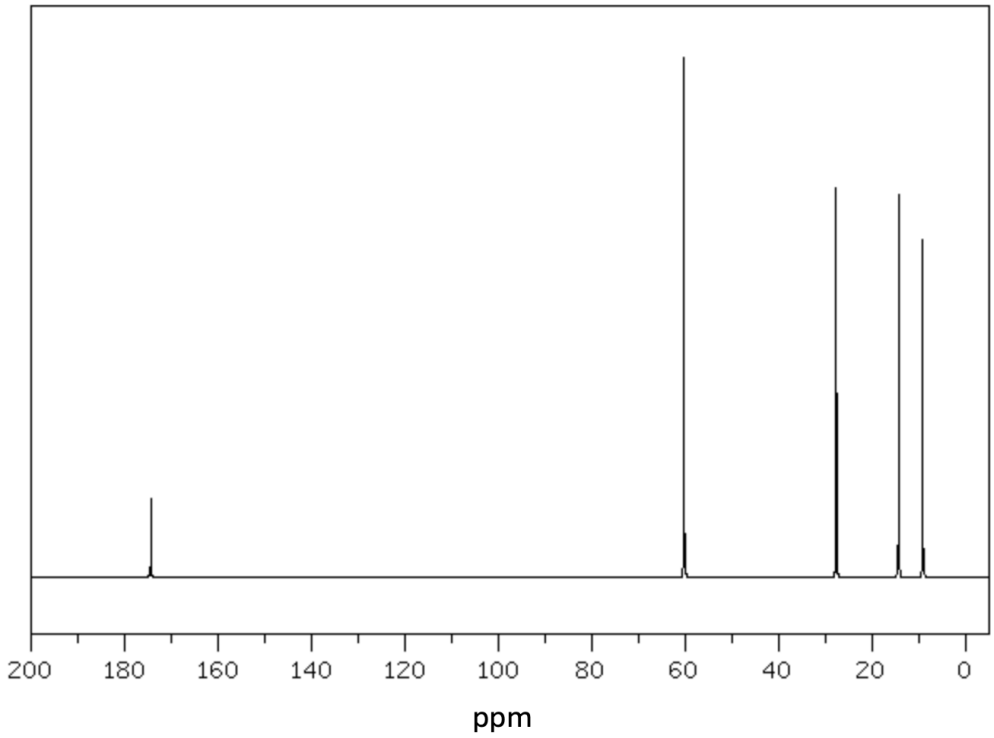
The order that the amino acids will be eluted is

A. Phenylalanine, serine, valine
B. Valine serine, phenylalanine
C. Serine, valine, phenylalanine
D. Phenylalanine, valine, serine

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Question 16

The ^{13}C NMR spectrum for a compound is shown below.



Source: https://www.chemicalbook.com/SpectrumEN_105-37-3_13CNMR.htm

Refer to **Item 22 of the Data Book.** Which of the following compounds is likely to have produced the spectrum above?

A. Butanamide

B. Pentanal

C. 3-methylbutanoic acid

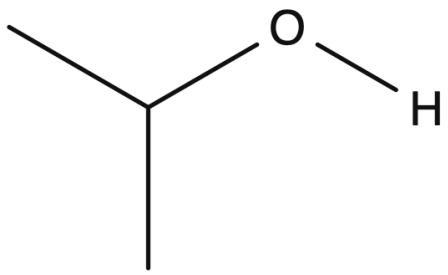
D. Ethyl propanoate

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Question 17

The skeletal structure of an organic compound is shown below.



Which of the following correctly describes the data that would be observed on the ^{1}H NMR spectrum for the above compound?

	Number of peak sets	Peak splitting	Relative peak areas
A.	4	1 singlet, 2 doublets, 1 septet	1:1:3:3
B.	4	1 singlet, 1 doublet, 2 quartets	1:1:3:3
C.	3	1 singlet, 1 doublet, 1 septet	1:1:6
D.	3	1 singlet, 1 doublet, 1 septet	1:1:2

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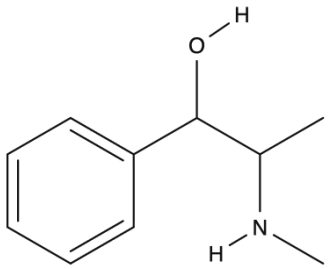
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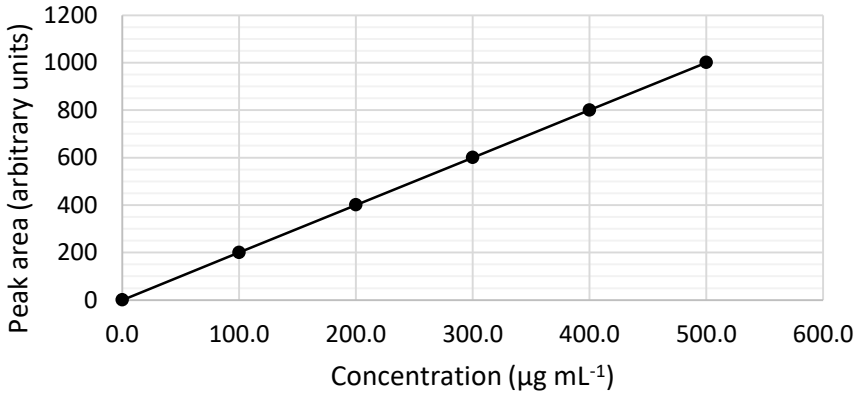
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Use the following information to answer Questions 18 and 19

Pseudoephedrine is a compound that is used to treat nasal congestion. The chemical structure of pseudoephedrine is shown below.



The calibration curve from the HPLC analysis for a series of pseudoephedrine standards of known concentration is shown below.



Question 18

The number of chiral centres in a pseudoephedrine molecule is

A. 0

B. 1

C. 2

D. 3

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Question 19

To determine the concentration of a liquid pseudoephedrine formulation, a 1.0 mL sample of the medicine is diluted to 20.0 mL, and a 0.10 mL aliquot is added to the mobile phase of the HPLC column under the same conditions as was used to construct the calibration curve above. The peak area obtained was 300 arbitrary units. The concentration of pseudoephedrine in the liquid medicine is closest to

A. $0.3\ mg\ mL^{-1}$
B. $1.5\ mg\ mL^{-1}$
C. $3.0\ mg\ mL^{-1}$
D. $7.5\ mg\ mL^{-1}$

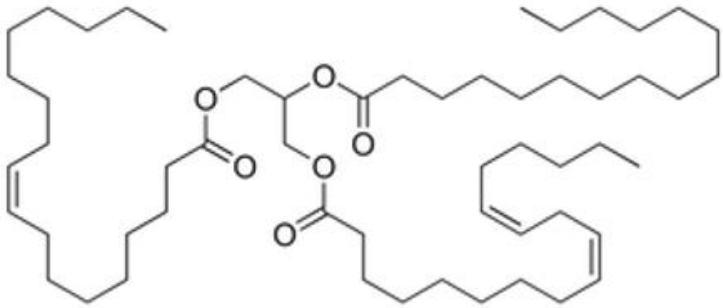
Question 20

Consider the process of cellular respiration and refer to **Item 12 of the Data Book**. At SLC, the maximum amount of energy, in kJ, obtained from cellular respiration in the presence of 20.0 mL of oxygen is closest to

A. $2.95 \times 10^{-1}\ kJ$
B. $3.82 \times 10^{-1}\ kJ$
C. $1.37 \times 10^{1}\ kJ$
D. $3.81 \times 10^{2}\ kJ$

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Use the following information to answer Questions 21 and 22.
The chemical structure of a triglyceride found in rice bran oil is shown below. Refer to the chemical structure and **Item 18 of the Data Book**.



https://www.researchgate.net/figure/The-fatty-acid-compositions-in-the-plant-derived-triglycerides-canola-oil-rice-bran-oil_fig2_339249336

Question 21
The triglyceride undergoes a transesterification reaction with methanol, in the presence of a catalyst. Which of the following would **not** be a product of this reaction?

A. $CH_3(CH_2)_{12}COOCH_3$
B. $CH_3(CH_2)_{14}COOCH_3$
C. $CH_3(CH_2)_7CHCH(CH_2)_7COOCH_3$
D. $CH_3(CH_2)_4CHCHCH_2CHCH(CH_2)_7COOCH_3$

Question 22
Which of the following represents the molecular formulas of the compounds produced from the enzyme-catalysed hydrolysis of the above triglyceride that occurs during digestion?

A. $C_3H_8O_3, C_{14}H_{28}O_2, C_{18}H_{32}O_2, C_{18}H_{34}O_2$
B. $C_3H_8O_3, C_{16}H_{30}O_2, C_{18}H_{34}O_2, C_{18}H_{32}O_2$
C. $C_3H_8O_3, C_{16}H_{32}O_2, C_{18}H_{36}O_2, C_{18}H_{32}O_2$
D. $C_3H_8O_3, C_{16}H_{32}O_2, C_{18}H_{32}O_2, C_{18}H_{34}O_2$

Question 23
Refer to **Item 19 of the Data Book**. Glutamic acid and asparagine are in a solution with a pH of 8.0. The charge of the ions formed by each of the amino acids, respectively are

A. $+1\ and\ +2$
B. $-1\ and\ -1$
C. $-2\ and\ -1$
D. $-1\ and\ -2$

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Question 24

The iodine value of a fat is defined by the mass of iodine, I_2, that reacts completely with 100 g of the substance.

A particular fatty acid has a molar mass of 276.0 g mol^{-1}, and an iodine value of 368. The number of carbon-carbon double bonds present in the compound is

A. 2

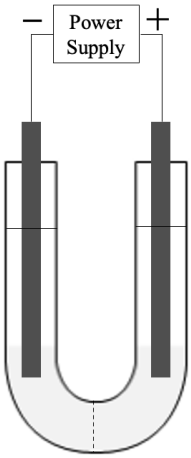
B. 4

C. 6

D. 8

Question 25

A student performed an experiment to investigate the electrolysis of an aqueous solution. Methyl red indicator was added to a 1.0 M solution of iron (II) sulfate in a U-tube. Two carbon electrodes were added to the solution, and connected to a 6.0 V power supply, as shown below.



Which of the following describes an expected observation as the cell operates?

A. Bubbles form at the surface of the negative electrode.

B. Solid metal forms at the surface of the positive electrode.

C. The size of the negative electrode decreases.

D. The solution changes colour from yellow to red at the positive electrode.

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Use the following information to answer Questions 26 and 27.

In acidic conditions, iron ions react with permanganate ions according to the chemical equation below.

$$5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(l)$$

A redox titration is performed to determine the concentration of iron (Fe^{2+}) present in a sample of a liquid iron supplement.

A 25.00 mL aliquot of the liquid supplement is titrated against a standardised solution of acidified potassium permanganate, $KMnO_4$ which has a concentration of 0.0105 M.

The titre volumes obtained from four trials are shown below.

Trial	Titre volume (mL)
1	10.20
2	9.95
3	10.00
4	9.90

Question 26

Based on the experiment data recorded, the %(m/v) of iron in the supplement is

A. 0.0233% (m/v)
B. 0.0291% (m/v)
C. 0.117% (m/v)
D. 0.146% (m/v)

Question 27

The actual concentration of iron (Fe^{2+}) in the sample is lower than the value determined using experimental data. Which of the following methods performed during the analysis is likely to contribute to this inaccurate result?

A. Rinsing the pipette with deionised water before delivering the aliquot of iron supplement.
B. Rinsing the conical flask with deionised water before adding the iron supplement aliquot.
C. Rinsing the burette with deionised water before adding the acidified potassium permanganate.
D. Adding the potassium permanganate from the burette to the aliquot of iron supplement before adding an indicator.

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Question 28

The recharge half-equations that occur in a nickel-metal hydride cell are shown below.

$M(s) + H_2O(l) + e^- \rightarrow MH(s) + OH^-(aq)$

$Ni(OH)_2(s) + OH^-(aq) \rightarrow NiOOH(s) + H_2O(l) + e^-$

The half-equation that occurs at the positive electrode as the cell provides electrical energy is

A. $M(s) + H_2O(l) + e^- \rightarrow MH(s) + OH^-(aq)$

B. $MH(s) + OH^-(aq) \rightarrow M(s) + H_2O(l) + e^-$

C. $Ni(OH)_2(s) + OH^-(aq) \rightarrow NiOOH(s) + H_2O(l) + e^-$

D. $NiOOH(s) + H_2O(l) + e^- \rightarrow Ni(OH)_2(s) + OH^-(aq)$

Question 29

Aluminium can be extracted via the electrolysis of molten aluminium oxide. If the cell operates with an average current of 300 kA, what mass of aluminium, in kg, can be produced in an hour?

A. 101 kg

B. 151 kg

C. 302 kg

D. 907 kg

Question 30

The value of the equilibrium constant for the ionisation of ethanoic acid at 25°C is $1.8 \times 10^{-5}\ M$. The reaction can be represented by the chemical equation shown below.

$$CH_3CH_2COOH(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CH_3CH_2COO^-(aq)$$

At equilibrium, the hydronium ion concentration in a 10 mL sample of ethanoic acid is $3.8 \times 10^{-3}\ M$. What is the concentration of ethanoic acid in the equilibrium mixture?

A. $1.2 \times 10^{0}\ M$

B. $8.0 \times 10^{-1}\ M$

C. $2.4 \times 10^{-3}\ M$

D. $4.7 \times 10^{-3}\ M$

End of Section A

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SECTION B - Short-answer questions

Instructions for Section B

- Answer all questions in the space provided.
- Write responses in English.
- Give simplified answers to all numerical questions, with an appropriate number of significant figures; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions; no marks will be awarded for an incorrect answer unless it is accompanied by working.
- Ensure all chemical equations are balanced and that the formulas for individual substances include an indication of state.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Question 1 (17 marks)

A student planned an experiment to investigate whether the amount of energy released upon combustion differs between structural isomers of fuels. The experiment plan from their logbook is shown below.

Aim

To determine whether the structure of an alcohol impacts the amount of energy released upon combustion.

Method

1. Measure out 100 mL deionised water using a measuring cylinder and add to the conical flask.
2. Attach the conical flask to the retort stand and adjust to a height so that the spirit burner can sit ~5 cm below the conical flask.
3. Clamp the thermometer above the flask, so it can remain submerged in the water throughout the reaction. Record the initial temperature of the water.
4. Add butan-1-ol to the spirit burner.
5. Weigh the spirit burner using an electronic balance and record the initial mass of the spirit burner and alcohol.
6. Place the spirit burner under the flask and adjust the height as needed. Light the spirit burner, and continuously stir the water, and monitor the temperature of the solution.
7. Once the temperature of the water has increased to ~50 degrees, extinguish the flame by replacing the cap of the spirit burner.
8. Continue to monitor the temperature, until the maximum temperature is reached, and record this temperature.
9. Once the spirit burner has cooled, reweigh the spirit burner and alcohol, and record the final mass.
10. Repeat Steps 1-9 with butan-2-ol and 2-methylpropan-2-ol.

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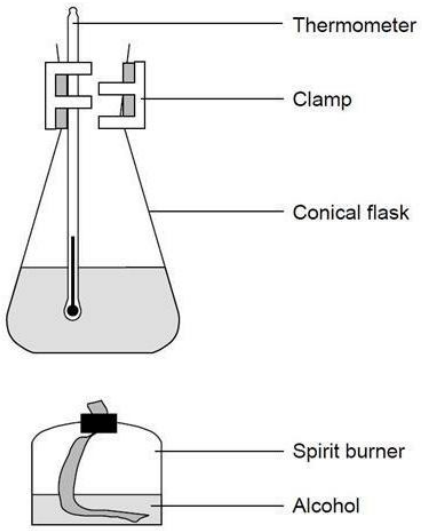


Figure 1. Diagram of experiment set up.
source: https://edu.rsc.org/experiments/comparing-heat-energy-from-burning-alcohols/1733.article

Results

		Temperature of water (°C)		Mass of spirit burner and alcohol (g)	
Alcohol	**Trial**	**Initial**	**Final**	**Initial**	**Final**
Butan-1-ol	**1**	23.5	52.0	186.04	184.76
	2				
	3				
Butan-2-ol	**1**				
	2				
	3				
2-methylpropan-2-ol	**1**				
	2				
	3				

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a. Identify the independent and dependent variable of the investigation.

2 marks

The student had to access the safety data sheets (SDS) for each of the alcohols prior to performing the experiment method.

b. Identify a hazard statement and precautionary measure likely to be present on the SDS for the alcohols.

1 mark

Based on the chemical structure of butan-1-ol, it is classified as a primary alcohol.

c. Draw the structural formula of 2-methylpropan-2-ol, and classify it as a primary, secondary or tertiary alcohol according to its structural features.

2 marks

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d. Describe one laboratory chemical test, and expected qualitative observations, that could be used to distinguish 2-methylpropan-2-ol from butan-1-ol.

2 marks

The table in the results section of the logbook shows the data recorded for the combustion of butan-1-ol.

e. Write a balanced chemical equation for the complete combustion of butan-1-ol at SLC.

1 mark

f. Based on the experimental data recorded for Trial 1, calculate the molar enthalpy of combustion of butan-1-ol.

3 marks

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The molar enthalpy of combustion of butan-1-ol according to literature is $-2676\ kJ\ mol^{-1}$.

g. Determine the efficiency of energy transfer to the water for the method followed using results from trial 1.

1 mark

The student records the following observations in relation to the combustion of 2-methylpropan-2-ol.

'Black soot forms at the bottom of the conical flask throughout the combustion reaction'.

h. Describe this observation using a balanced chemical equation to represent the relevant process. Explain the impact this will have on the molar enthalpy of combustion calculated for 2-methylpropan-2-ol using experimental data.

3 marks

i. Discuss the method choice to include three trials for each of the alcohols in evaluating the quality of the data.

2 marks

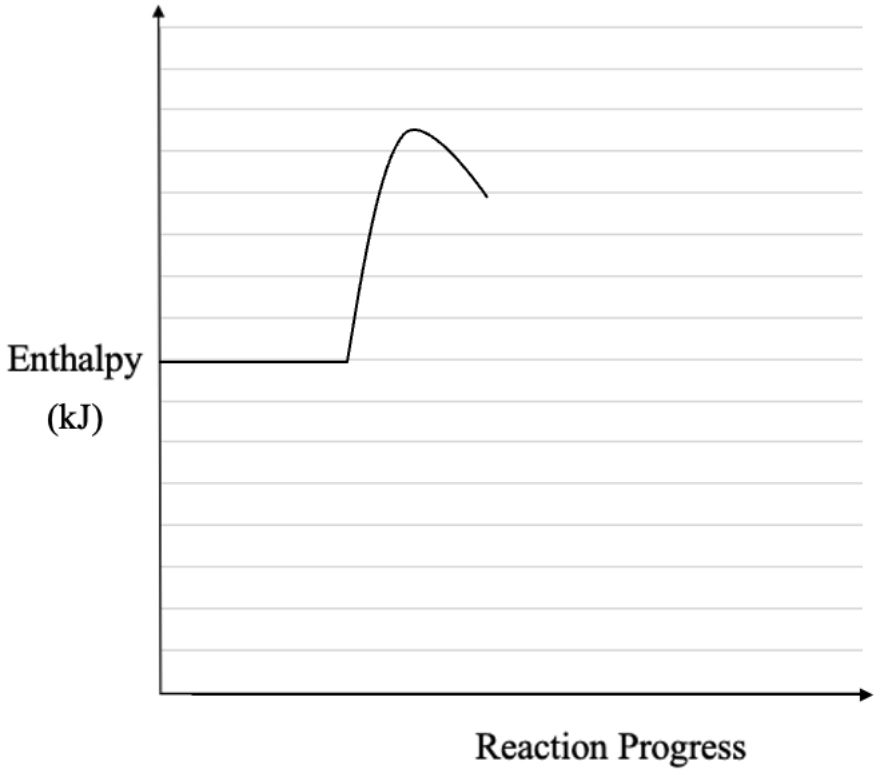
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Question 2 (15 marks)

Ethanol may be produced from ethene obtained from crude oil, as represented by the chemical equation shown below.

$$CH_2CH_2(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g) \quad \Delta H = -45\ kJ$$

An incomplete energy profile diagram for the reaction is shown below. The activation energy for the reaction is 135 kJ.



a. Complete the energy profile diagram above for the reaction between ethene and water to produce ethanol. Label the value of ΔH.

2 marks

A phosphoric acid catalyst is often used in this reaction.

b. Annotate the diagram above to show how the presence of a catalyst affects the energy profile diagram for the reaction.

1 mark

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At 520 K the value of the equilibrium constant, K_c, for the reaction is $1.48 \times 10^{-1}\ M^{-1}$.

c. Interpret the value of the equilibrium constant at this temperature.

1 mark

At equilibrium at 520K, the concentration of water is $5.11\ M$ and the concentration of ethene is $1.18\ M$.

d. Calculate the concentration of ethanol when the system is at equilibrium.

1 mark

The phosphoric acid catalyst allows for an intermediate reaction temperature of 520 K to be used in the production of ethanol.

e. In considering the rate and yield of ethanol production, explain the role of the catalyst and the reaction temperature of 520 K.

2 marks

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f. Assuming the reaction conditions remain as outlined above, apply Le Chatelier's principle to explain how the yield of ethanol could be increased at a temperature of 520 K.

3 marks

The following methods may also be used to produce ethanol.

Feedstock	**Description of reaction to produce ethanol**
Crude oil	Multi-step synthesis via the substitution reactions of ethane
Plant biomass	Fermentation of glucose

g. With reference to relevant green chemistry principles, compare and evaluate the three methods of ethanol production.

5 marks

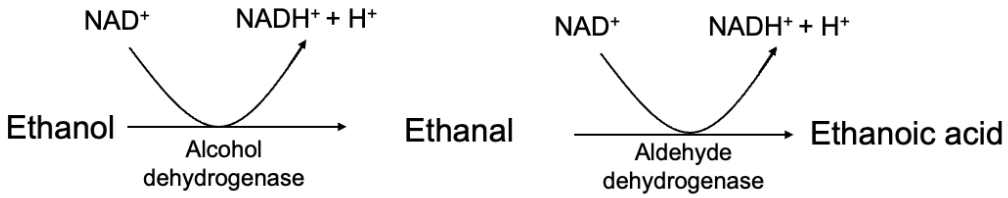
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Question 3 (5 marks)

Alcohol dehydrogenases and aldehyde dehydrogenases are classes of enzymes involved in the metabolism of alcohols and aldehydes in the body.

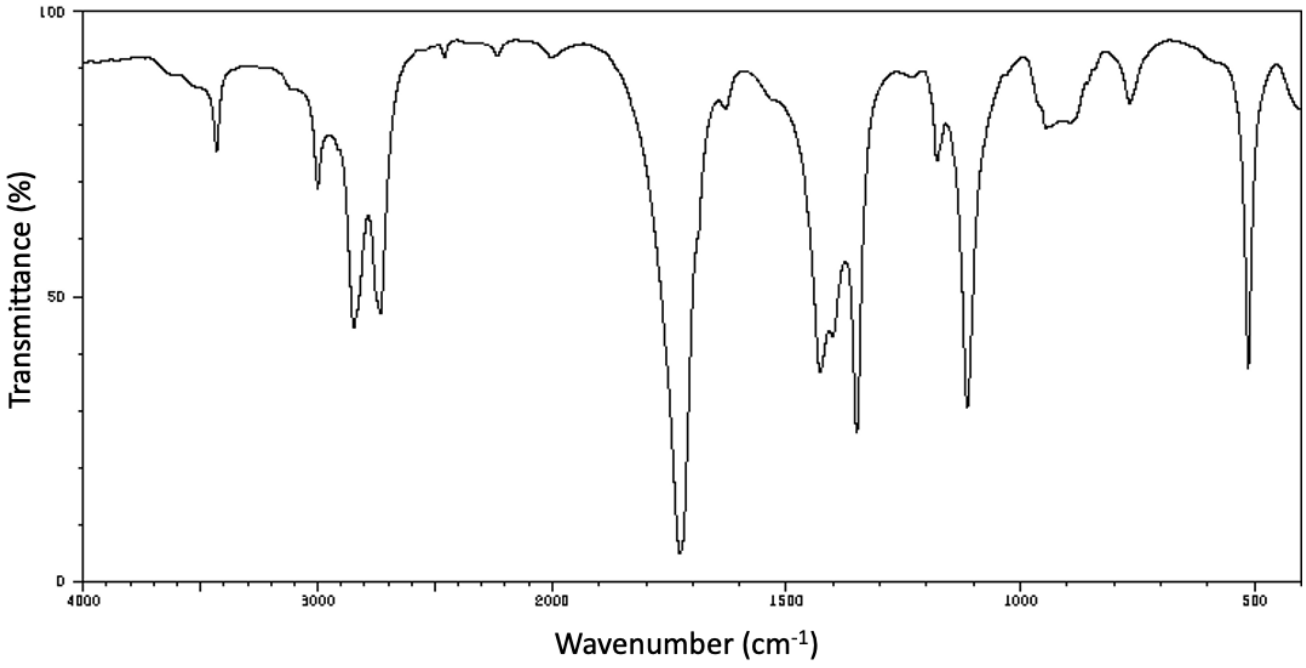
Ethanol, or drinking alcohol, is metabolised according to the diagram shown below.



To test the activity of the dehydrogenases at different temperatures, the enzymes were added to an ethanol solution at varying temperatures. At varying time points, samples of the reaction mixture were tested for the presence of metabolites.

The following spectral data was obtained from the analysis of samples taken at different time points following the addition of enzymes at a given temperature.

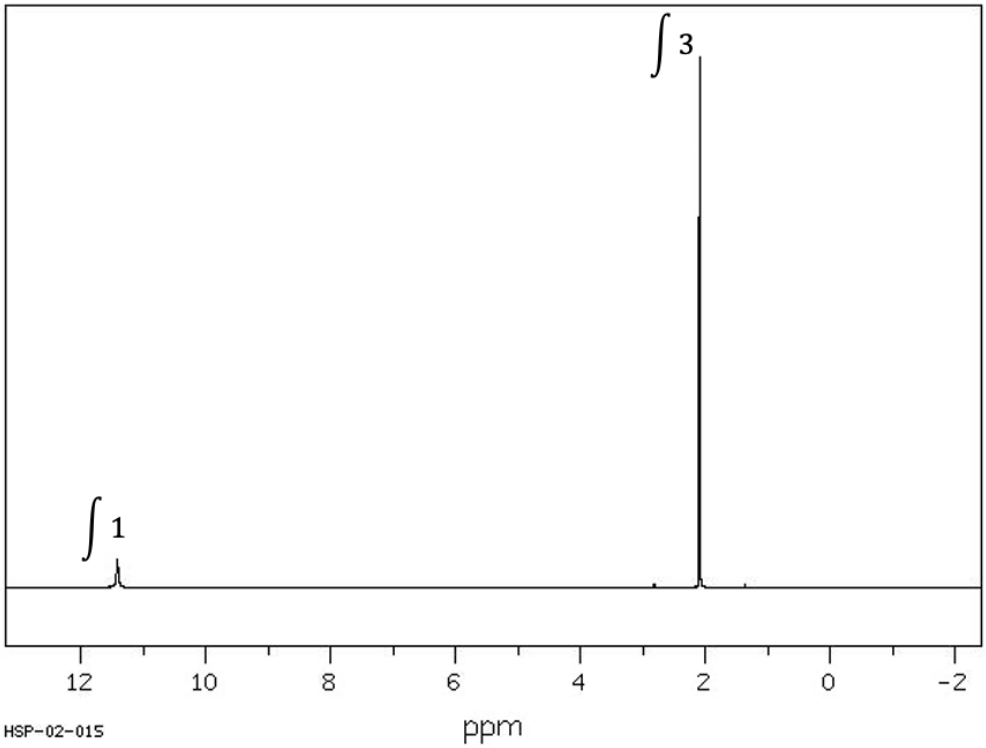
Infrared spectrum of a compound present in the sample.



https://www.chemicalbook.com/SpectrumEN_75-07-0_IR1.htm

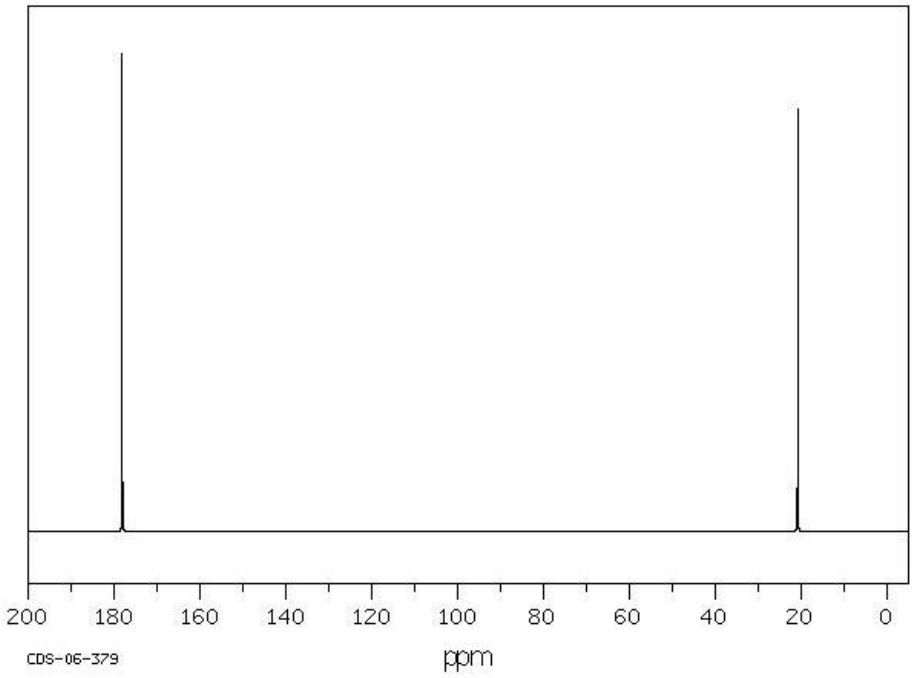
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^{1}H NMR spectrum of a compound present in the sample.



https://www.chemicalbook.com/SpectrumEN_64-19-7_1HNMR.htm

^{13}C NMR spectrum of a compound present in the sample.



https://www.chemicalbook.com/SpectrumEN_64-19-7_1HNMR.htm

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Referring to relevant data provided in the three spectra, explain the conclusions that may be drawn about the activity of the dehydrogenases at the temperature that the sample was obtained from.

5 marks

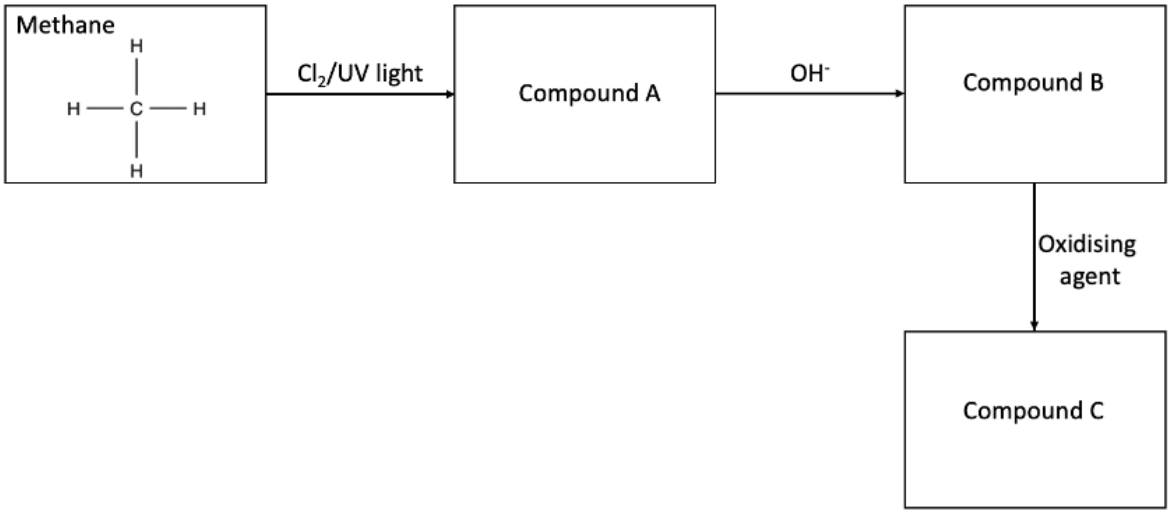
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Question 4 (17 marks)

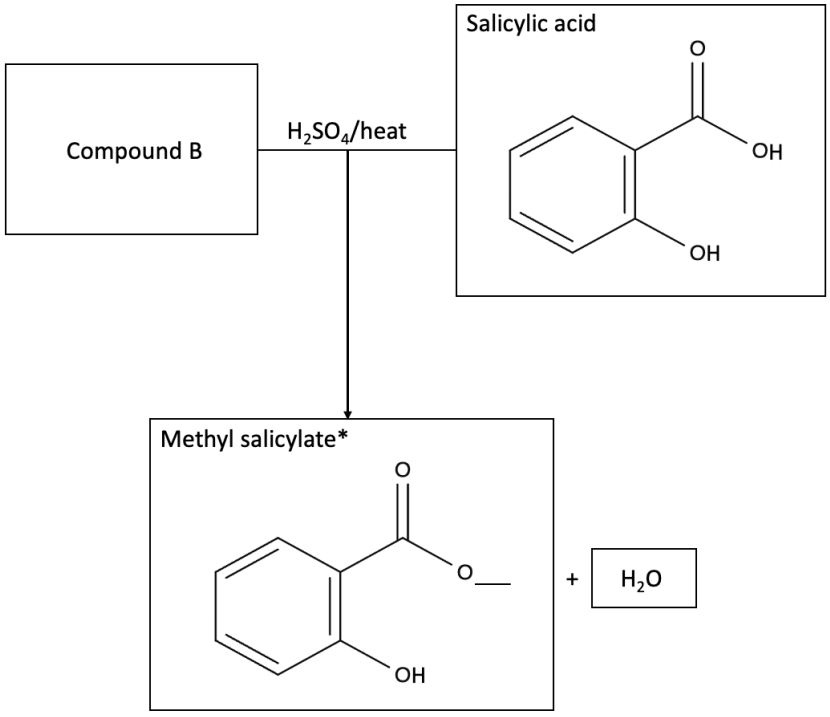
The synthesis of Compound B from methane is outlined in **Reaction Pathway 1**.

Reaction Pathway 1



Methyl salicylate is a fragrant compound commonly found in topical pain relief medications used in the treatment of muscle aches. Methyl salicylate is produced from salicylic acid, and Compound B, as shown in the reaction pathway below.

Reaction Pathway 2



***Formula for methyl salicylate is incomplete**

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Refer to **Reaction Pathway** 1 on the previous page. In the first step of this reaction pathway, Compound A is produced from methane.

a. Provide the IUPAC name of Compound A and classify the type of chemical reaction that occurs.

2 marks

b. Provide the IUPAC name of Compound B.

1 mark

Compound B may also be produced from Compound C, in the presence of a reducing agent, as shown in Reaction Pathway 1.

c. Draw the structural formula of Compound C in the space below.

1 mark

Consider the analysis of Compound B and Compound C using infrared spectroscopy.

d. Refer to **Item 21 of the Data Book** to compare the infrared spectrum produced from the analysis of these two compounds.

3 marks

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Refer to **Reaction Pathway 2** on the previous page. The structural formula of methyl salicylate shown in the diagram is incomplete.

e. In the space provided in the box labelled methyl salicylate in **Reaction Pathway 2**, complete the structure shown by writing the formula of the missing group in the space provided.

1 mark

The reaction represented in reaction pathway 2 exists as an equilibrium mixture.

f. Write a balanced chemical equation, including states, to represent the equilibrium system, using the molecular formulas of compounds.

2 marks

g. Write the expression for the equilibrium constant, K.

1 mark

Distillation apparatus may be used to separate the components of the equilibrium mixture.

h. Describe how fractional distillation allows for the separation of compounds from a reaction mixture.

2 marks

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i. With reference to structure and bonding, explain the separation of the reaction mixture using factional distillation.

4 marks

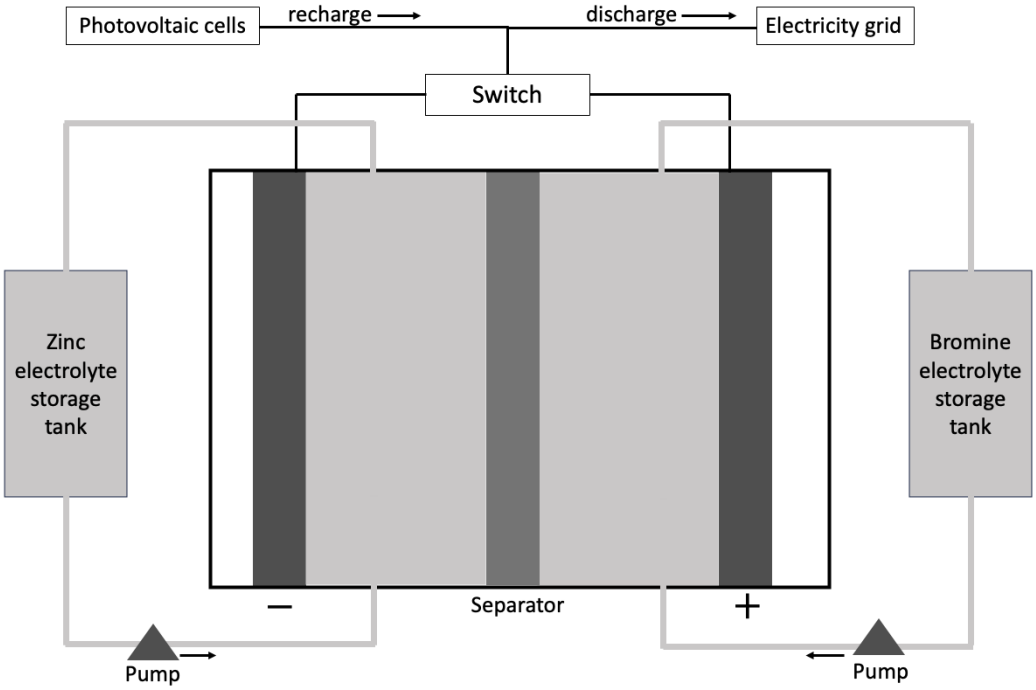
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Question 5 (11 marks)

Redox flow batteries have been researched to provide large-scale energy storage for excess electrical energy generated from renewable sources, such as wind and solar.

The diagram below shows a zinc-bromine redox battery, that stores energy generated by photovoltaic cells, which can then be used to provide electricity when required.



An acidified solution of zinc bromide is used as the electrolyte, and the separator allows for the selective movement of zinc and bromide ions. As the cell charges, solid zinc is deposited on the surface of the negative electrode.

The overall reaction as the cell provides electricity is represented by the chemical equation below.

$$Br_2(aq) + Zn(s) \rightarrow Zn^{2+}(aq) + 2Br^-(aq)$$

a. State the energy transformation that occurs as the cell discharges.

1 mark

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b. Refer to **Item 2 of the Data Book** to determine the maximum voltage produced by a zinc-bromine cell, assuming SLC.

1 mark

Consider the reaction that occurs as energy from the solar cells is provided.

c. Write balanced half-equations for the reactions that occur at the anode and the cathode as energy is provided to the cell.

2 marks

Anode:

Cathode:

d. Describe the function of the separator in the zinc-bromine cell.

2 marks

The cell has a discharge time of 10 hours, delivering an average continuous current of 105 A.

e. Calculate the mass of zinc, in kg, that reacts at the negative electrode during a discharge cycle.

3 marks

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Zinc-bromine redox flow cells typically have an operating efficiency of around 70%.

f. Describe two features of the negative electrode that would be considered in the design of an efficient zinc-bromine cell.

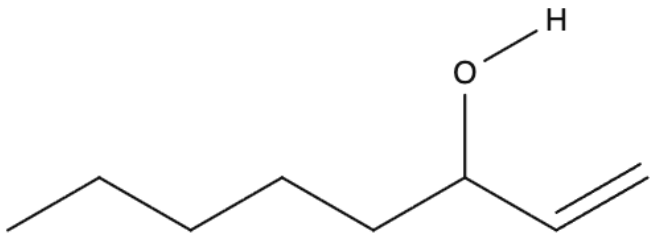
2 marks

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Question 6 (9 marks)

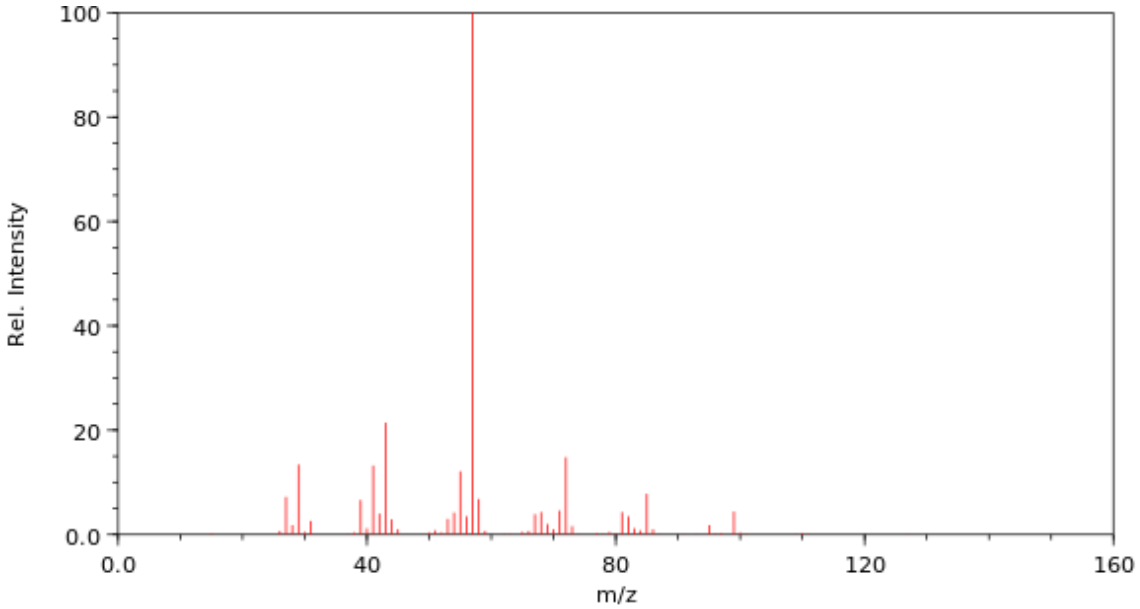
The chemical compound shown below is found in human sweat and breath and it is thought to play a role in attracting mosquitos.



a. Write the IUPAC name for the compound.

1 mark

The mass spectrum for the compound is shown below.



https://webbook.nist.gov/cgi/cbook.cgi?ID=C3391864&Mask=200

b. Write the formula of the fragment ion that produces the base peak.

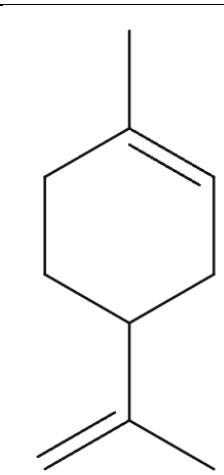
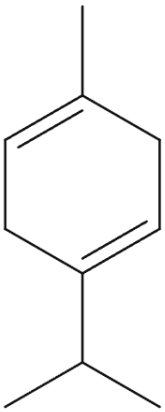
1 mark

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c. Provide the semi-structural formula of a structural isomer that does **not** contain the same functional groups as the compound shown on the previous page.

1 mark

The chemical structures of limonene and terpinene, two compounds commonly used in natural mosquito repellents, are shown below. These compounds are contained in eucalyptus oil, which is extracted from eucalyptus leaves using steam distillation. The distillate obtained contains a layer of water and oil.

	
limonene	terpinene

Solvent extraction is an alternative method that may be used to extract active compounds from eucalyptus leaves.

d. Select and describe the features of an appropriate solvent that would allow for the extractions of compounds such as limonene and terpinene from eucalyptus leaves.

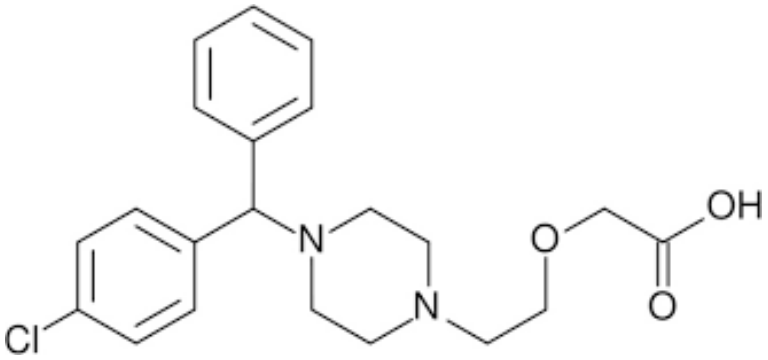
2 marks

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Antihistamine medication, such as cetirizine, can be used to treat the itching associated with mosquito bites. Cetirizine has biological activity through forming hydrogen bonds and dipole-dipole interactions with histamine receptors.

The chemical structure of cetirizine is shown below.



Source: researchgate.net/figure/Chemical-structure-of-cetirizine_fig1_258703936

Cetirizine is classified as an acid.

e. Identify the functional group that contributes to the acidic properties of cetirizine.

1 mark

Cetirizine contains a chiral centre, and its synthesis produces two enantiomers.

f. Circle the chiral centre on the structure of cetirizine shown above.

1 mark

Only one of the cetirizine enantiomers is effective as an antihistamine medication.

g. Explain the differences in biological activity of the two enantiomers of cetirizine.

2 marks

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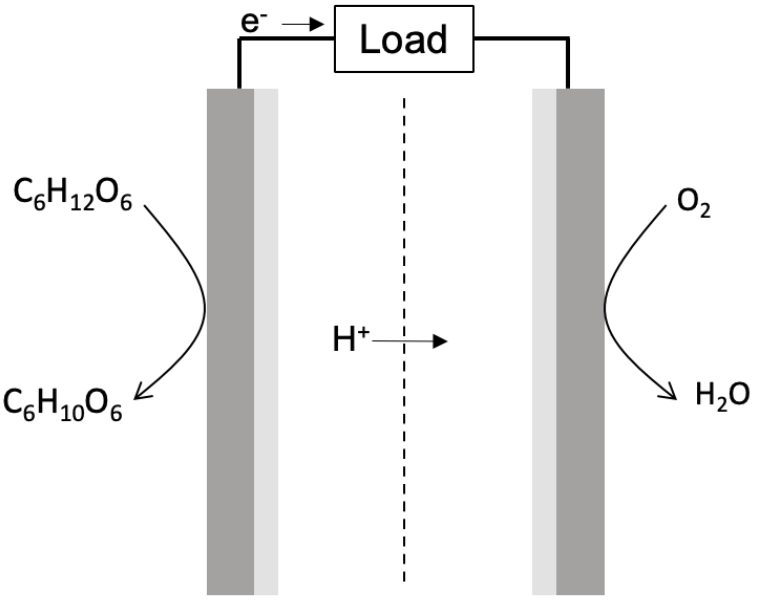
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Question 7 (16 marks)

Enzyme fuel cells are a type of biofuel cell that utilise enzymes at the electrodes to facilitate the cell reactions. A particular enzyme fuel cell utilises glucose as the fuel. A diagram of the fuel cell, and an overview of one of the reaction stages that occurs as the cell is in operation is shown below.



Research into the use of these fuel cells for medical implants has been explored, given the availability of glucose in the body.

a. With reference to relevant chemical reactions, describe the storage of glucose in the body.

2 marks

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Glucose is a major energy source for the body. In aerobic respiration, 1.0 g of glucose releases 15.57 kJ of energy.

b. Use this information to write a balanced thermochemical equation to represent the aerobic cellular respiration.

3 marks

Refer to the diagram of the glucose enzyme fuel cell.

c. Write a balanced chemical equation for the reaction that occurs at the anode as the cell operates.

1 mark

The cell produces a voltage of 0.80 V.

d. Use this information to determine the standard electrode potential of glucose at SLC.

1 mark

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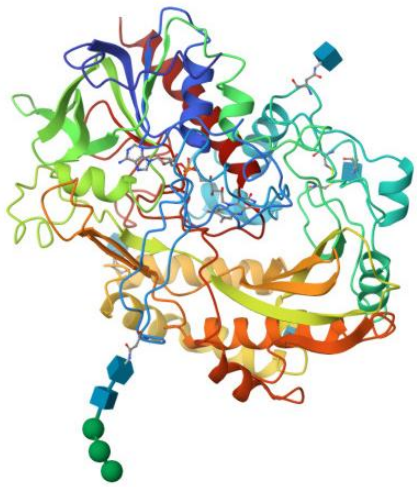
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The cell is found to operate with an efficiency of 40%.

e. Assuming the cell delivers a constant current of $300\ \mu A$, calculate the amount of time, in days, taken for the cell to consume 1.0 g of glucose.

3 marks

One of the enzymes utilised in this fuel cell design is glucose oxidase. The crystal structure of glucose oxidase is shown below.



Source: https://www.rcsb.org/structure/1GAL

The tertiary structure of an enzyme refers to the overall three-dimensional shape.

f. Describe the bonding that determines the tertiary structure of a protein.

2 marks

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g. Discuss the design and operating features of the glucose enzyme fuel cell that should be considered in ensuring that the activity of glucose oxidase is maintained as the cell operates.

4 marks

End of Section B

End of examination questions

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Extra space for responses

Clearly number all responses in this space.

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